

Telecom CDR ETL Pipeline

Hackathon Use Case — Informatica IICS

Project Overview

Build an ETL pipeline using Informatica Cloud (IICS) to process daily Telecom Call Detail Records (CDR). The pipeline extracts raw CSV data, cleans and transforms it, loads it into a database, and generates KPI reports for revenue and usage analysis.

Prerequisites	Tools Required
Understand telecom terms: duration, call types (Local/S	D/INT), tower IDsInformatica IICS — Extract, Transform, Load
Knowledge of ETL pipeline concepts	Database — staging tables and final tables
Basic SQL for KPI queries	Sample CDR CSV file

System Architecture

Source CDR CSV → Informatica Extract → Informatica Transform → Database (Staging + Final) → KPI Dashboard



Daily CDR Source Expr/Filter/ stg_cdr Revenue CSV File Qualifier Lookup final_cdr KPIs

Step	Action	Informatica Component
1	Extract daily CDR CSV file from storage	Source Qualifier
2	Remove duplicates, fix dates, replace NULLs	Expression + Filter + Sorter
3	Calculate Revenue & IsInternational fields	Expression Transformation
4	Load into stg_cdr then final_cdr	Target Designer + Lookup
5	Generate KPI reports for all 4 use cases	Aggregator + SQL Queries

Core Features (Must-Have)

- ✓ **Extract**
 - Load a daily CDR CSV file from a storage location
 - Filter: only process current day's records
- ✓ **Transform**
 - Remove duplicate CDR entries

- Fix incorrect or inconsistent date/time formats
- Replace missing call durations with 0
- Revenue = Duration × 0.02 (charge per second)
- IsInternational = 1 if CallType = 'INT', else 0

✓ Load

- Load cleaned data into database staging table (stg_cdr)
- Load final data into production table (final_cdr)

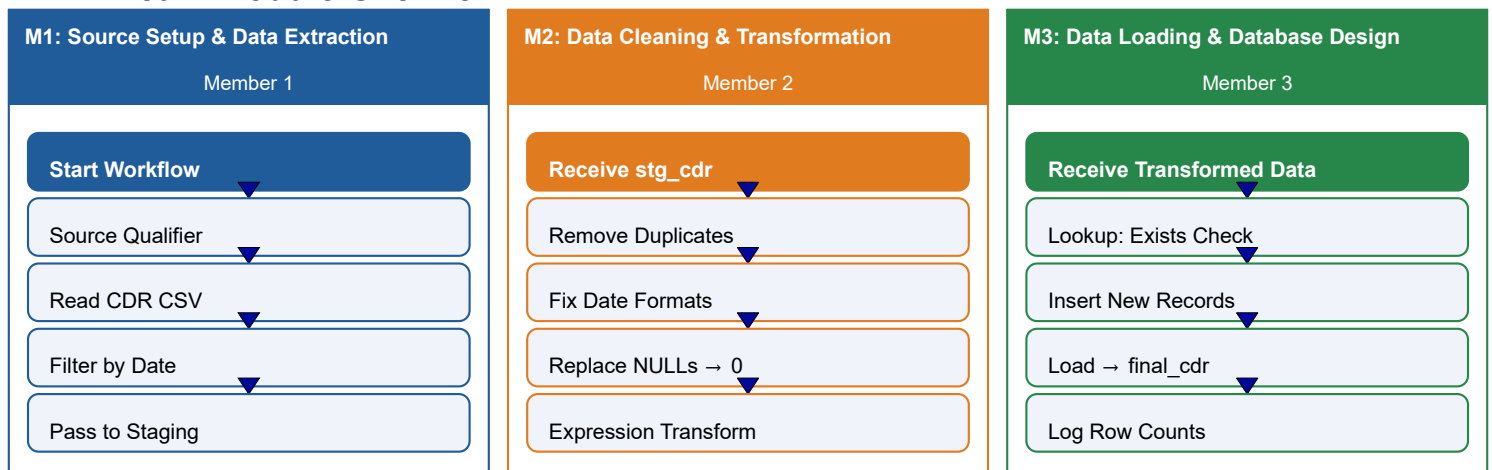
Stretch Features (Enhanced)

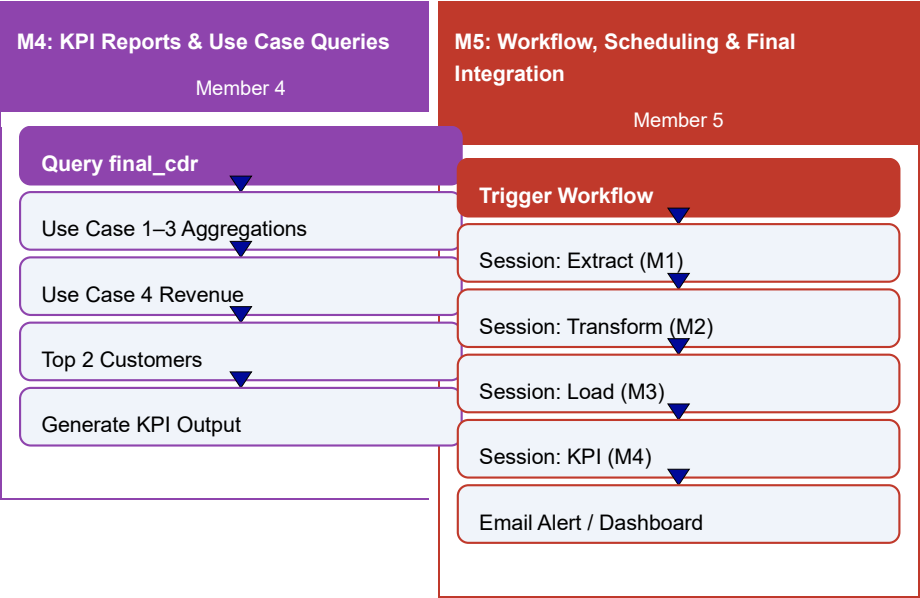
- ✓ **Dashboard** — Build a Customer Usage Dashboard using final data (Excel report)
- ✓ **Email Alert Notification** — Send success/failure email after each pipeline run
- ✓ **Incremental Load** — Load only new/changed records each day
- ✓ **Auto-retry** — Retry failed sessions up to 3 times automatically

KPI Use Cases — Final Output

Use Case	Category	KPIs / Metrics
Use Case 1	Call Analysis	Total calls per customer Calls by type: Local / STD / INT
Use Case 2	Duration Analysis	Total call duration per customer Average call duration
Use Case 3	Tower Analysis	Total calls per tower Most used tower identification
Use Case 4	Revenue Data	Revenue per customer % Contribution to total revenue Top 2 customers by revenue

Team Module Overview





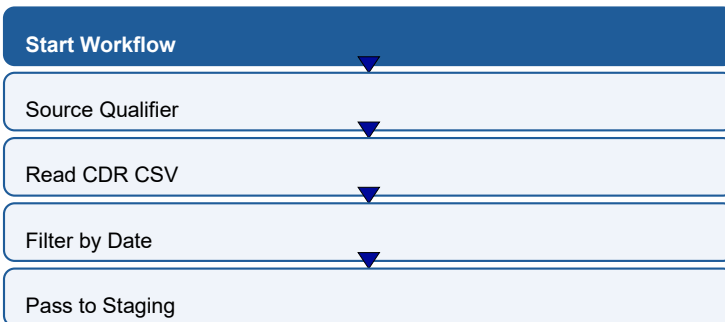
Objective

Set up the source connection to the daily CDR CSV file and extract raw call records into the Informatica staging environment using Source Qualifier.

Design Phase

- Identify CDR CSV file structure: CustomerID, CallType, Duration, CallDate, TowerID
- Configure Source Analyzer in Informatica IICS
- Create source definition for CSV flat file
- Design extraction filter: load only today's CDR file (daily batch)
- Map source columns to staging table columns

Workflow



Implementation

- Informatica: Source Analyzer → import CSV schema
- Source Qualifier transformation with date filter
- Staging table: stg_cdr (CustomerID, CallType, Duration, CallDate, TowerID)
- Schedule: daily batch trigger at midnight
- Log row count on each successful extraction

Stretch Features

- Schedule daily automated extraction
- Log extraction row counts and file names

Output

Raw CDR records loaded into stg_cdr staging table — ready for transformation

Objective

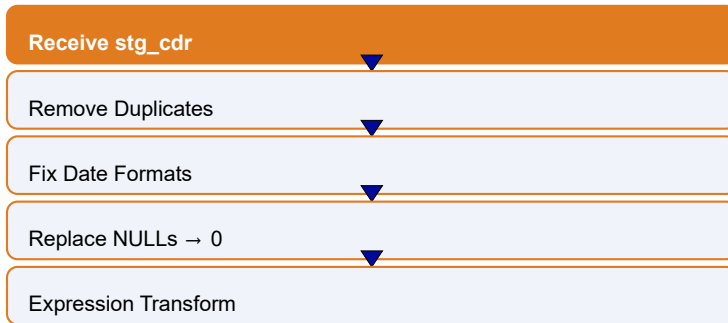
Clean the raw CDR data by removing duplicates, fixing date/time formats, replacing missing call durations, and applying all required transformation rules using Informatica.

Design Phase

- Identify duplicate CDR records (same CustomerID + CallDate + Duration)

- Detect and fix inconsistent date/time formats (e.g., DD-MM-YYYY vs MM/DD/YYYY)
- Define rule: Replace NULL call durations with 0
- Design Expression Transformation for calculated fields
- Design Router to separate valid vs. invalid records

Workflow



Implementation

- Sorter + Aggregator: Identify and remove duplicate rows
- Expression: IIF(ISNULL(Duration), 0, Duration)
- Expression: Revenue = Duration × 0.02
- Expression: IsInternational = IIF(CallType = 'INT', 1, 0)
- Router: route invalid records to error table, valid to next step

Stretch Features

- Log all rejected/invalid records with reason
- Reprocess failed records pipeline

Output Cleaned, transformed CDR data with Revenue and IsInternational fields populated

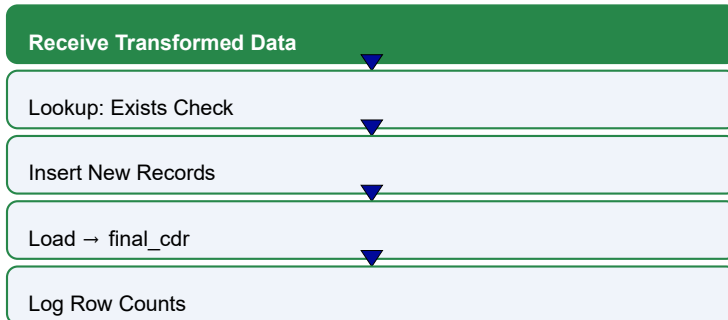
Objective

Design and create the final database tables, load the cleaned and transformed CDR data from staging into the final production table ready for KPI reporting.

Design Phase

- Design staging table schema: stg_cdr
- Design final table schema: final_cdr with all computed fields
- Plan incremental load strategy (avoid reloading existing records)
- Design Lookup to check if record already exists before insert
- Plan error table: err_cdr for rejected records

Workflow



Implementation

- Create DB tables: stg_cdr, final_cdr, err_cdr
- Final table columns: CustomerID, CallType, Duration, Revenue, IsInternational, CallDate, TowerID
- Lookup Transformation: check if CustomerID+CallDate already loaded
- Target Designer: configure final_cdr as Informatica target
- Load rejected records into err_cdr with error reason

Stretch Features

- Incremental load (only new records)
- Audit table with load timestamps

Output

final_cdr table fully populated with cleaned, enriched CDR data — ready for KPI queries

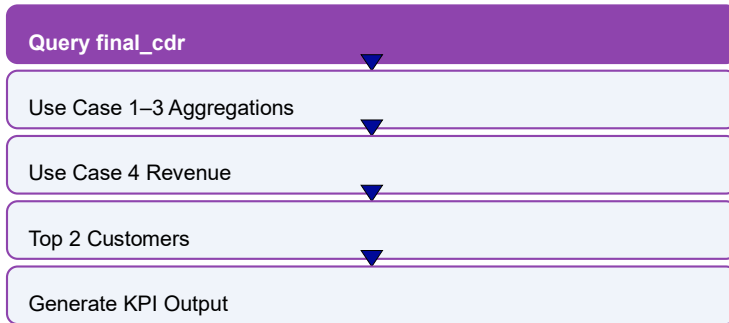
Objective

Build KPI queries and reports from the final_cdr table covering all 4 use cases: Call Analysis, Duration Analysis, Tower Analysis, and Revenue Data.

Design Phase

- Use Case 1 — Call Analysis: Total calls per customer, calls by type (Local/STD/INT)
- Use Case 2 — Duration Analysis: Total duration per customer, average call duration
- Use Case 3 — Tower Analysis: Calls per tower, most used tower
- Use Case 4 — Revenue Data: Revenue per customer, % contribution, Top 2 customers• Plan output format: SQL views or Excel export for dashboard

Workflow



Implementation

- KPI 1: SELECT CustomerID, COUNT(*) AS TotalCalls, CallType FROM final_cdr GROUP BY CustomerID, CallType
- KPI 2: SELECT CustomerID, SUM(Duration) AS TotalDuration, AVG(Duration) AS AvgDuration FROM final_cdr GROUP BY CustomerID
- KPI 3: SELECT TowerID, COUNT(*) AS TotalCalls FROM final_cdr GROUP BY TowerID ORDER BY TotalCalls DESC
- KPI 4: SELECT CustomerID, SUM(Revenue) AS TotalRevenue, SUM(Revenue)*100.0/SUM(SUM(Revenue)) OVER() AS PctContribution FROM final_cdr GROUP BY CustomerID
- Top 2: ORDER BY TotalRevenue DESC LIMIT 2

Final KPI Output

Use Case	Category	KPIs / Metrics
Use Case 1	Call Analysis	Total calls per customer by type (Local/STD/INT)
Use Case 2	Duration Analysis	Total & average call duration per customer
Use Case 3	Tower Analysis	Calls per tower, most-used tower identification
Use Case 4	Revenue Data	Revenue per customer, % contribution, Top 2 customers

Stretch Features

- Export KPIs to Excel dashboard
- Auto-alert for Top revenue customer changes

Output

4 KPI report tables / SQL views covering Call, Duration, Tower, and Revenue analysis

Module 5: Workflow, Scheduling & Final Integration — Member 5

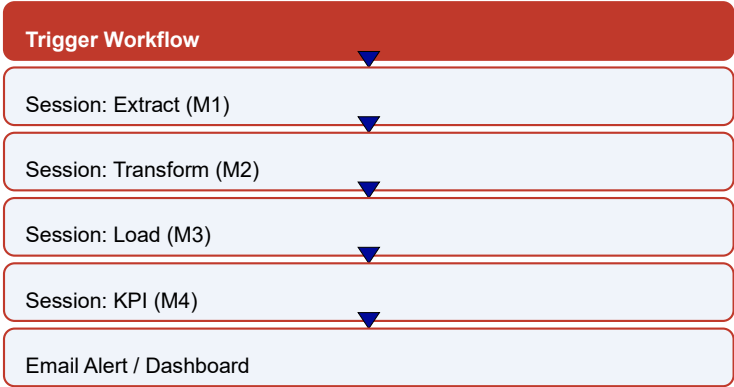
Objective

Integrate all 4 modules into a single end-to-end Informatica workflow, configure scheduling, error handling, email alert notifications, and build the Customer Usage Dashboard.

Design Phase

- Design end-to-end workflow: Extract → Transform → Load → KPI
- Link all session tasks (M1→M2→M3→M4) in sequence
- Design error handling: reject into err_cdr, log with reason codes
- Plan email alert notification on pipeline failure or completion
- Design Customer Usage Dashboard layout (Excel report)

Workflow



Implementation

- Workflow Manager: Create master workflow with 4 session tasks
- Command Task: pre-check CSV file exists before extraction
- Error handling: err_cdr table + failure log file
- Scheduler: Daily at midnight (cron or Informatica scheduler)
- Email Service: Send success/failure notification after each run
- Stretch: Build Customer Usage Dashboard in Excel with Revenue + KPI data

Stretch Features

- Customer Usage Dashboard (Excel report with charts)
- Email Alert Notification on failure/success

Integration Strategy

Module	Member	Input From	Output To
M1: Extract	Member 1	CDR CSV File	stg_cdr table
M2: Transform	Member 2	stg_cdr table	Cleaned records
M3: Load	Member 3	Cleaned records	final_cdr table
M4: KPI	Member 4	final_cdr table	KPI reports / views
M5: Workflow	Member 5	All sessions (M1–M4)	Automated pipeline + Dashboard

End-to-End Workflow

1. CDR CSV file arrives daily

Output	Fully automated, monitored ETL pipeline running daily with email alerts and dashboard
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2. M1: Extract raw data → stg_cdr
3. M2: Clean & transform data (Revenue, IsInternational)
4. M3: Load into final_cdr production table
5. M4: Run KPI queries → Use Cases 1–4
6. M5: Monitor run + send email alert + update dashboard

Team Collaboration Guidelines

- Use shared GitHub repository for all mappings and scripts
- Follow naming conventions: stg_cdr, final_cdr, err_cdr
- Test extractions and transformations using Postman / Workflow Monitor
- Communicate table schema changes with the full team before implementing
- Use Informatica Workflow Monitor to check session logs daily